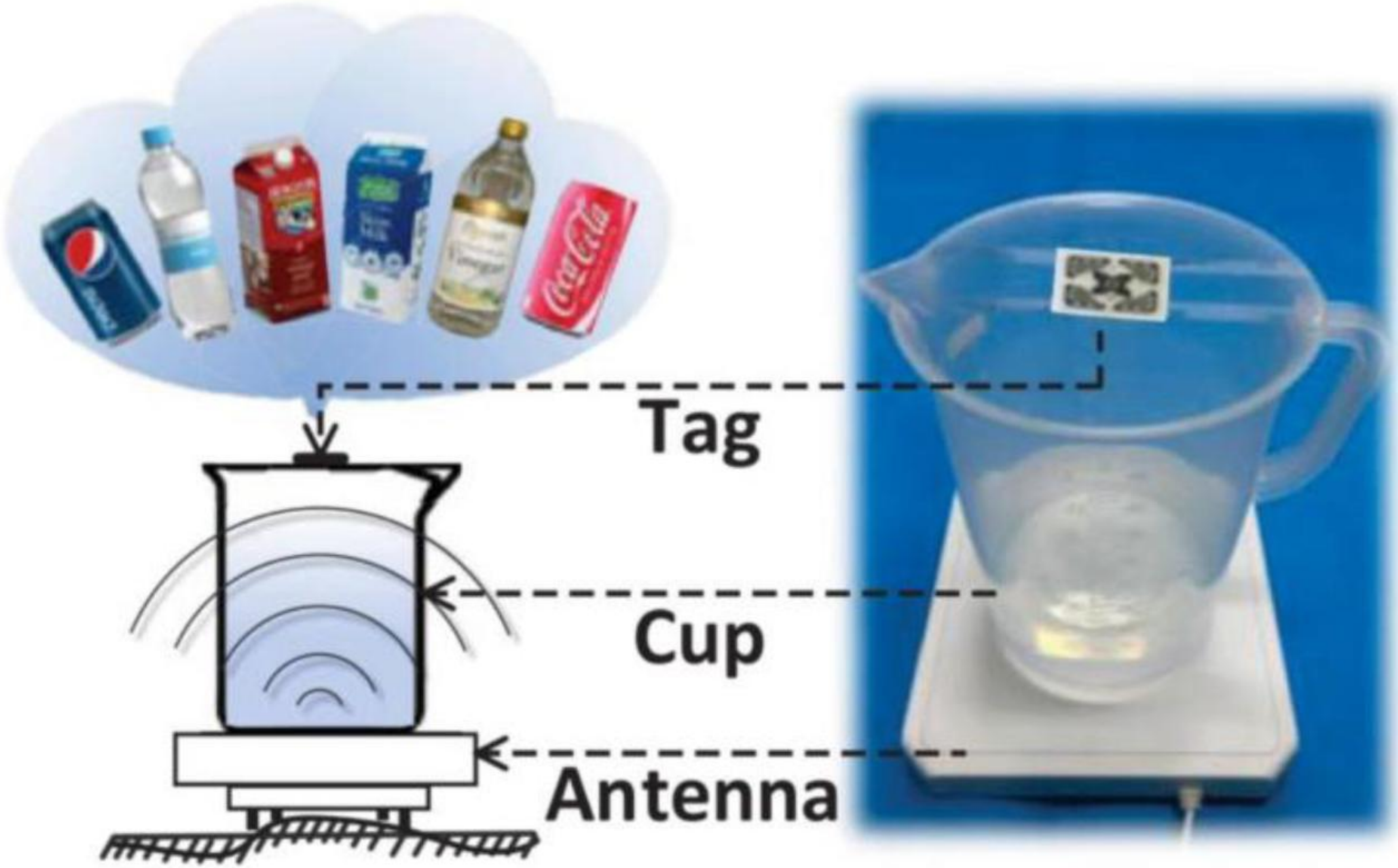
工科创2-智能物联网

课程project实操讲解-1

2026.5

Project2 基于反向散射标签的液体识别



Pluto SDR的基本控制代码

创建SDR对象

```
%% Initialize SDR device
deviceNameSDR = 'Pluto'; % Set SDR Device
radio = sdrdev(deviceNameSDR); % Create SDR device object
```

定义发射波形及参数

```
%% Prepare wave
samp_rate = 10e6; %Sampling ratetxGain = 0;
txGain = 0;
rxGain = 10;
% if_freq = 50e3; % Intermediate frequency
frameLen = 5e6; % Frame length in num. of samples
t = 0:1/samp_rate:(frameLen-1)/samp_rate; % Discrete time sequence
% txWave = exp(1j*2*pi*if_freq*t).';
txWave = ones(length(t),1)+1j*ones(length(t),1);
```

直接通信

加标签反射

定义发射发射机、接收机

```
%% Transmitter set
sdrTransmitter = sdrtx(deviceNameSDR); % Transmitter properties
sdrTransmitter.RadioID = 'usb:0';
sdrTransmitter.BasebandSampleRate = samp_rate;
% sdrTransmitter.CenterFrequency = 2.412e9; % Channel 1 of Wi-Fi
sdrTransmitter.CenterFrequency = 915e6; % Frequency of backscatter tag
sdrTransmitter.ShowAdvancedProperties = true;
sdrTransmitter.Gain = txGain;
```

```
%% Receiver set
sdrReceiver = sdrrx(deviceNameSDR);
sdrReceiver.RadioID = 'usb:0';
sdrReceiver.BasebandSampleRate = samp_rate;
sdrReceiver.CenterFrequency = sdrTransmitter.CenterFrequency;
sdrReceiver.GainSource = 'Manual';
sdrReceiver.Gain = rxGain;
sdrReceiver.OutputDataType = 'double';

captureLen = frameLen;
sdrTransmitter.transmitRepeat(txWave);
burstCaptures = capture(sdrReceiver, captureLen, 'Samples');
```

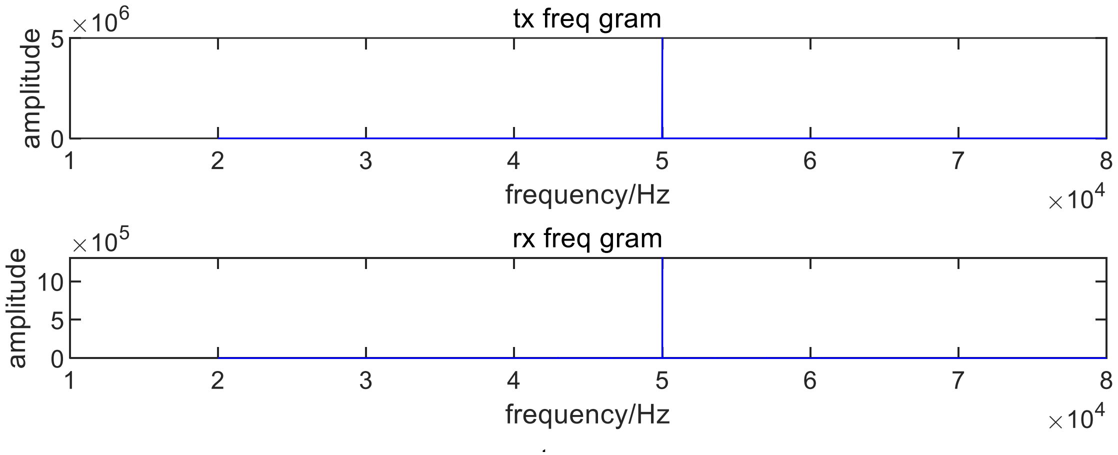
每个函数的具体使用，可以使用help命令查看MATLAB的官方文档

例如：

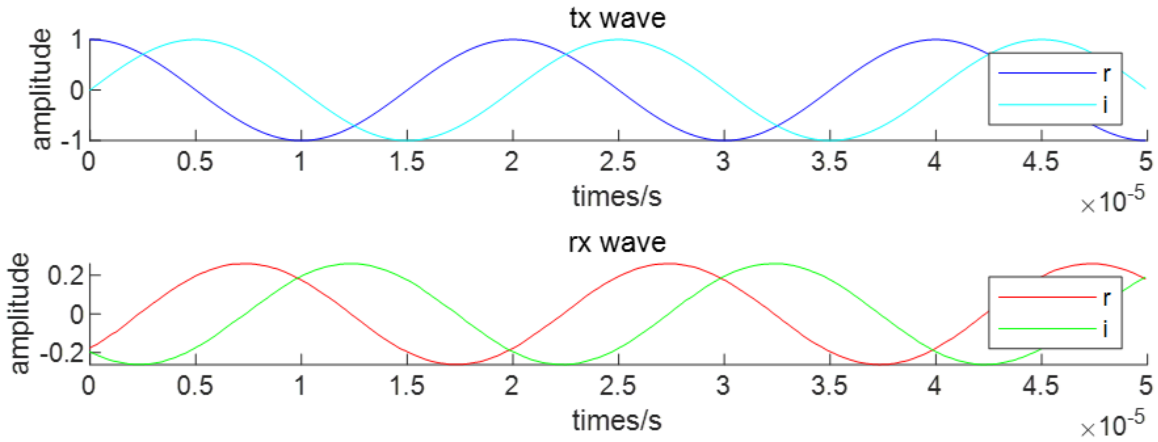
```
>> help capture
--- comm.SDRRxPluto/capture 的帮助 ---
```

信号如何分析?

收到的信号为复数信号，即有两个维度，可以从时域、频域、**IQ平面**分析



频域

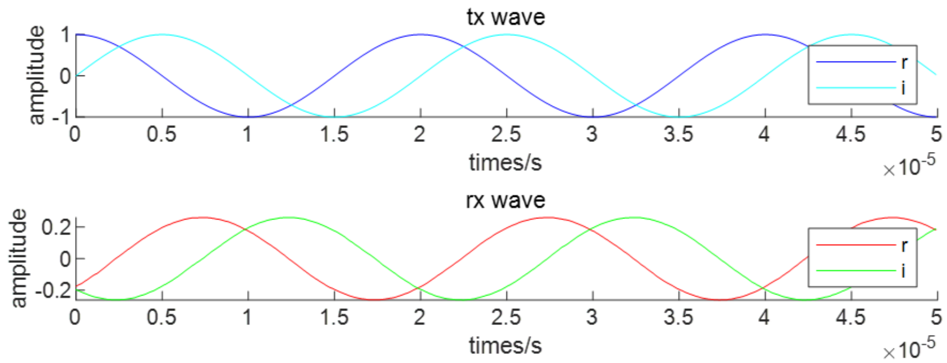


时域

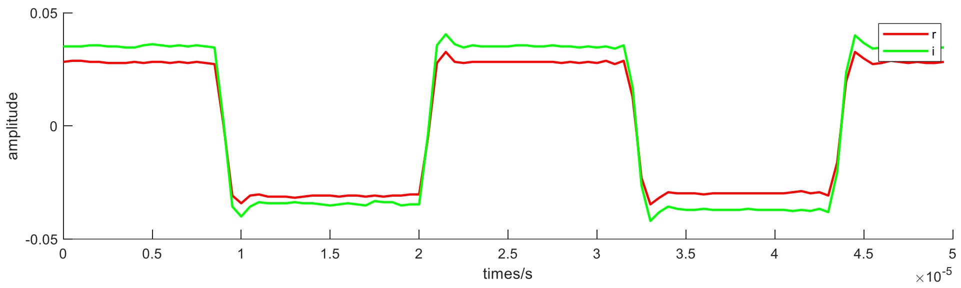
testPlutoSDR.m 的 %% data plot 部分包含了如何从时域、频域观察信号

测试：直接通信与反射

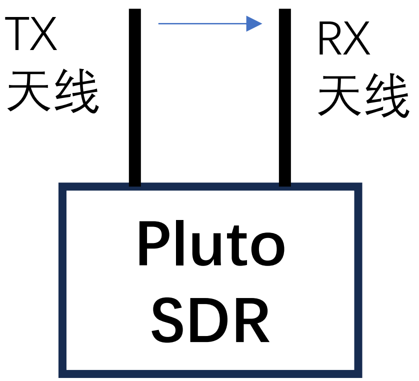
直接通信（以发射正弦信号为例）

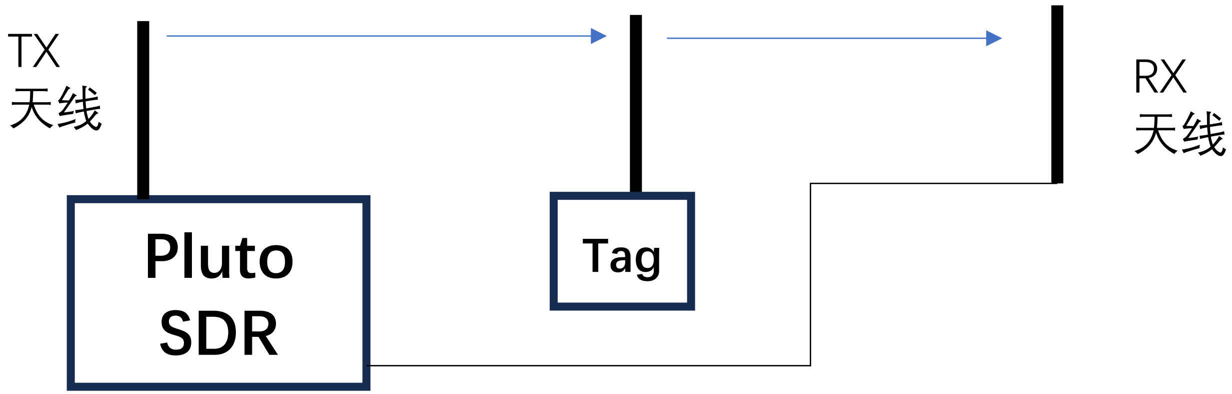


加标签反射



直接通信:





testPlutoSDR.m 的 %% data plot 部分包含了如何从时域、频域观察信号

不同天线

注：这里的频率是载波频率，
不是基带频率
因此更换天线时应该对应修改：

```
%% Transmitter set
sdrTransmitter = sdrtx(deviceNameSDR); % Transmitter properties
sdrTransmitter.RadioID = 'usb:0';
sdrTransmitter.BasebandSampleRate = samp_rate;
% sdrTransmitter.CenterFrequency = 2.412e9; % Channel 1 of Wi-Fi
sdrTransmitter.CenterFrequency = 915e6; % Frequency of backscatter tag
sdrTransmitter.ShowAdvancedProperties = true;
sdrTransmitter.Gain = txGain;
```

信号如何分析?

从IQ平面上：**直接在复平面上画出收到的复数信号**